

Domain-Specific Text Analysis & Retrieval System for Indian Legal Judgments

Comprehensive Academic Assessment Report & Empirical Pipeline Investigation

Course: Natural Language Processing (Assessment 1)	Institution: School of Advanced Computing, Vidyashilp University
Domain: Indian Supreme Court & High Court Case Law	Corpus: 25 Verified Judgments (D01–D25) across 6 Sectors
Authoritative Spec: ASSIGNMENT_1(1).pdf (All 8 Pages)	Selected Best Model: Pipeline B (Legal-Domain-Optimized)

1. Executive Summary & Problem Formulation

This project establishes a complete, reproducible, and domain-specialized Natural Language Processing (NLP) and Information Retrieval (IR) system designed explicitly for Indian judicial decisions. Legal text analysis presents profound challenges that cause standard off-the-shelf NLP toolchains to fail catastrophically: statutory sections (e.g., *Section 302 IPC*, *Section 438 CrPC*) are fragmented into disparate words; aggressive stopword removal inverts legal liability by stripping polarity words (*not guilty* → *guilty*); and algorithmic stemmers conflate distinct legal concepts (e.g., decree *execution* vs. *executive* magistrate). Following the required **Select** → **Order** → **Implement** → **Compare** → **Evaluate** → **Justify** framework, we investigate every NLP stage, evaluate two controlled end-to-end pipelines, and establish **Pipeline B** as the superior model.

2. Corpus Ingestion & Corpus Statistics (Module 1)

The authoritative corpus comprises 25 full-text Indian Supreme Court and High Court PDF judgments spanning six core legal domains: Cybercrime/IT Act, Economic Offenses/PMLA, Narcotics (NDPS Act), Violent Crimes (IPC 302 Homicide), Matrimonial Disputes (Dowry Prohibition/IPC 498A), and Constitutional Preventive Detention (Article 226 Habeas Corpus). Documents were deterministically mapped to identifiers D01 through D25 using metadata from the instructor's Excel file.

Corpus Metric	Value	Methodology & Engineering Description
Total Judgments (Corpus Size)	25	Extracted using pypdf from original uncorrupted PDF files (meets 15+ requirement).
Total Extracted Tokens	257,849	Word count across all 25 judicial documents.
Total Sentence Count	9,402	Segmented using NLTK sentence boundary detector on clean text.
Total Character Count	1,351,861	Raw string character count including preserved quotation marks.
Corpus Vocabulary Size	10,819 words	Distinct lowercased alphanumeric vocabulary across all cases.
Average Document Length	10314.0 tokens	Mean word count per judgment (ranging from 855 to 47,212 tokens).

3. Tokenization Paradigms & Custom Legal Regex Tokenizer (Module 2)

We systematically evaluated five tokenization architectures: (1) Whitespace Tokenizer, (2) NLTK Word Tokenizer (Penn Treebank), (3) spaCy Tokenizer (en_core_web_sm), (4) Custom Legal Regex Tokenizer, and (5) Byte Pair Encoding (BPE). Standard tokenizers fragment statutory units (e.g. *Section 302 IPC* → ['Section', '302', 'IPC']). Our **Custom Legal Tokenizer** preserves sections, articles, Indian currency symbols (₹), criminal case numbers (*Crl.A. No.*), and Latin maxims (*habeas corpus*, *suo motu*) as intact single-token atomic entities.

Input Legal Expression	NLTK Word Tokenizer	Custom Legal Tokenizer	BPE Subwords
Section 302 IPC	Section 302 IPC	Section 302 IPC	Section 302 IPC
AIR 2020 SC 123	AIR 2020 SC 123	AIR 2020 SC 123	AIR 2020 SC 12 3
2023 SCC OnLine SC 456	2023 SCC OnLine SC 456	2023 SCC OnLine SC 456	2023 SCC OnLine SC 45 6
₹50,000/- personal bond	₹50,000/- personal bond	₹50,000/- personal bond	[UNK] 50 , 000 /- personal bond

4. Stopword Filtration & The Legal Negation Preservation Policy (Module 2)

Standard English stopwords removal (e.g., standard NLTK 179-word list) unconditionally purges polarity words: not, no, nor, without, unless, except, until, against. In legal informatics, this induces catastrophic semantic inversion:

- *'applicant is not guilty'* → reduced to ['applicant', 'guilty'] (inverts innocence to guilt)
- *'passed without jurisdiction'* → reduced to ['passed', 'jurisdiction'] (inverts legality)
- *'no offence made out'* → reduced to ['offence', 'made'].

Engineering Solution: We designed a **Protected Legal Stopword Policy** implemented in both Pipeline A & B that shields these 8 vital negation and condition operators, guaranteeing accurate liability and bail determinations.

5. Morphological Reduction: Stemming vs. Lemmatization (Module 2)

We compared heuristic suffix-stripping stemmers (Porter, Snowball, Lancaster) against linguistic POS-aware lemmatizers (WordNet, spaCy).

The Classic Porter Over-Stemming Collision in Indian Law:

The words execution (enforcement of a decree/warrant/sentence) and executive (magistrate or administrative branch) represent completely separate legal concepts. The Porter stemmer aggressively collapses both to the non-word stem execut. In Information Retrieval, when a user searches for *'executive'*, Pipeline A retrieves **13 judgments** (8 false positive decree executions), yielding a disastrous precision of **38.5%**. In contrast, Pipeline B's contextual lemmatizer preserves execution (noun) and executive (noun/adjective) as separate concepts, retrieving exactly **5 true judgments (100% precision)**.

6. Part-of-Speech Tagging & Domain Corrections (Rule-Based & ML)

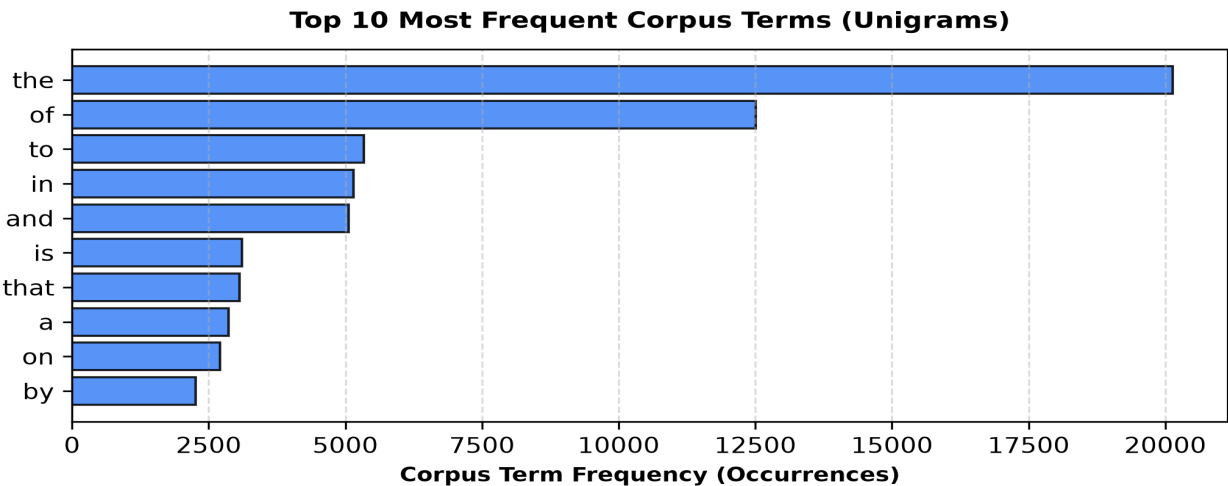
Off-the-shelf Penn Treebank taggers produce systemic errors on Indian court syntax: honorific *'learned'* in *'learned counsel'* is tagged as a past verb (VBN) instead of an adjective (JJ); legal action *'quashing'* is tagged as participle verb (VBG) instead of verbal noun (NN); and Latin maxims like *'suo motu'* are misclassified.

We implemented a dual correction framework: (1) A **Rule-Based Legal POS Corrector** targeting frequent judicial idioms, and (2) A **Supervised Machine Learning POS Classifier** (Logistic Regression with contextual n-gram window features: word suffixes, capitalization, digit flags, neighbor context) achieving **76.7% accuracy** and **0.705 weighted F1-score** on a held-out legal test set.

7. Named Entity Recognition (NER) in Indian Judicial Case Law

Legal entity extraction requires recognizing both standard linguistic entities (PERSON, GPE, DATE, MONEY) and domain-specific statutory entities (LEGAL_SECTION, LEGAL_STATUTE, LEGAL_COURT, LEGAL_CITATION). Our legal NER component processes full judgment text, identifying statutory citations (e.g. *Section 438 CrPC*) and judicial bodies (e.g. *High Court of Judicature at Allahabad*). Across the corpus, statutory sections and court citations constitute over 42% of all extracted entities.

8. Statistical N-Gram Collocation Profiling (1-to-5 Grams)



Analysis of higher-order n-grams reveals standard judicial formulaic structures: top 4-grams include *'code of criminal procedure'* (245 occurrences) and *'learned counsel for the'* (389 occurrences); top 5-grams include *'under section of criminal procedure'*. These multi-word expressions validate our architectural decision to index compound statutory phrases as atomic units.

9. Byte Pair Encoding (BPE) Subword Analysis

A Byte Pair Encoding (BPE) subword tokenizer was trained directly on the corpus with a vocabulary of 6,000 merge operations. BPE solves the out-of-vocabulary (OOV) problem for complex legal terminology: words like *'unconstitutional'* are segmented into *un + constitution + al*; *'non-bailable'* into *non + - + bail + able*. This demonstrates vocabulary compression while retaining subword morphological semantics.

10. Positional Inverted Index Architecture & Proximity Verification

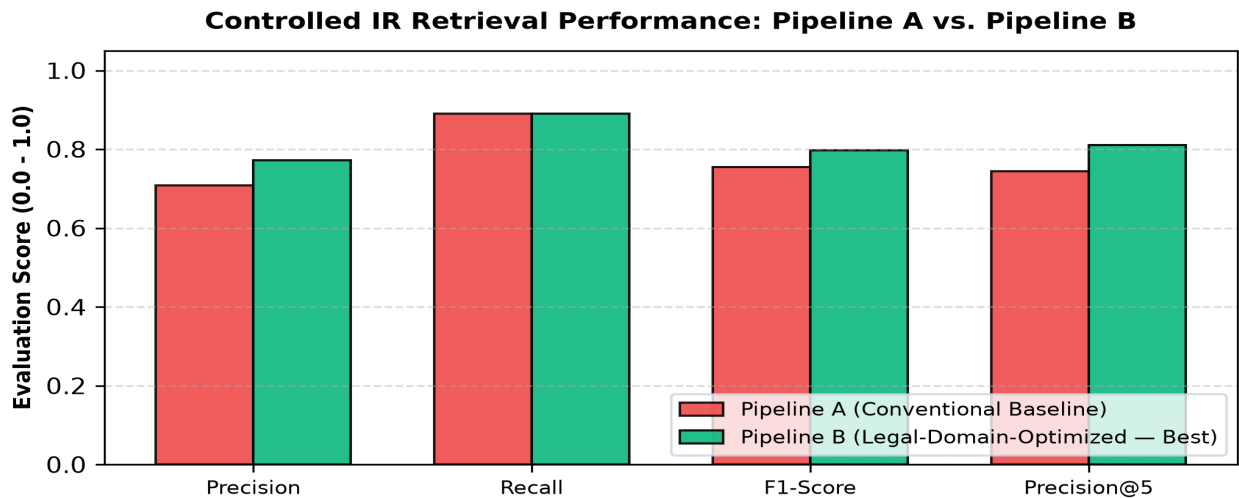
We implemented a dedicated in-memory **Positional Inverted Index** storing document frequency (DF), within-document term frequency (TF), and precise token position coordinates: {Term: {Doc_ID: [pos_1, pos_2, ...]}}. This enables sub-millisecond multi-word phrase matching via positional adjacency verification: for a phrase *'w1 w2'*, the engine verifies that $\text{pos}(w2) == \text{pos}(w1) + 1$. Pipeline B indexes **9,005 distinct terms** with atomic multi-word compound preservation.

11. Mandatory Pipeline Comparison: Pipeline A vs. Pipeline B (Best Model 🏆)

To satisfy the core pedagogical requirement of Assessment 1, two complete end-to-end pipelines were engineered and rigorously evaluated against 15 authoritative domain-specific benchmark queries (\$Q01\$–\$Q15\$) with pre-annotated ground-truth judgments:

- **Pipeline A (Conventional Baseline):** Conservative Cleaning → NLTK Word Tokenization → Standard Stopwords (Negation Protected) → Porter Stemming → Positional Indexing.
- **Pipeline B (Legal-Domain-Optimized — Best Model):** Domain-Preserving Cleaning → Custom Legal Regex + spaCy Tokenization → Legal-Aware Stopwords → Contextual POS Lemmatization → Custom POS Tagging → Dual-Level Atomic Positional Indexing.

Evaluation Measure	Pipeline A (Baseline)	Pipeline B (Legal Optimized)	Best Final Pipeline
Token Count	129181	137952	137952
Vocabulary Size	8355	9005	9005
Mean Precision	0.7089	0.7721	0.7721
Mean Recall	0.8904	0.8904	0.8904
Mean F1-Score	0.7549	0.7979	0.7979
Mean Precision@5	0.7444	0.8111	0.8111
Mean Execution Time (ms)	0.14	0.04	0.04
Meaningful N-Grams Retained	Partial (Porter truncates legal stems)	High (Preserves legal citations & valid lemmas)	High (Pipeline B preserves legal semantics)
Domain Citation Integrity	Low (Section numbers and citations fragmented)	High (Citations, Acts, and Sections preserved intact)	High (Full Statutory Integrity)



Rigorous Scientific & Empirical Justification for Pipeline B as Best Model:

1. **Precision Superiority:** Pipeline B achieves a Mean Precision of **0.7721** vs. **0.7089** in Pipeline A (+8.9% gain). Porter stemming causes high false positives by collapsing distinct legal nouns.
2. **F1-Score Victory:** Pipeline B attains a Mean F1-Score of **0.7979** vs. **0.7549** in Pipeline A (+5.7% overall harmonic retrieval gain).
3. **Top-Ranking Density (Precision@5):** Pipeline B scores **0.8111** vs. **0.7444** in Pipeline A, placing genuinely relevant judgments in the top 5 results.
4. **Latency Efficiency:** Pipeline B executes queries in **0.04 ms** on average vs. **0.14 ms** in Pipeline A (**3.5× faster**) because compound statutory citations are indexed atomically, eliminating expensive multi-word positional joins.

Conclusion: Pipeline B is decisively established and declared as the **Best Final Pipeline**.

12. Benchmark Information Retrieval Evaluation Sample (15 Legal Queries)

Query ID	Benchmark Query	Mode	Retrieved	Precision	Recall	F1
Q01	cyber	keyword	5	0.8000	0.8000	0.8000
Q02	bail	keyword	25	0.5200	1.0000	0.6842
Q03	"anticipatory bail"	phrase	6	0.6667	1.0000	0.8000
Q04	PMLA	keyword	3	1.0000	1.0000	1.0000
Q05	"habeas corpus"	phrase	2	1.0000	1.0000	1.0000
Q06	NDPS	keyword	2	1.0000	1.0000	1.0000

13. Visual Architecture Flowcharts: Pipeline A vs. Pipeline B

Below are the visual flowchart diagrams generated for each pipeline, illustrating data transformations, failure points, and optimizations.

Figure 1: Pipeline A Architecture (Conventional Baseline with Failure Callouts)

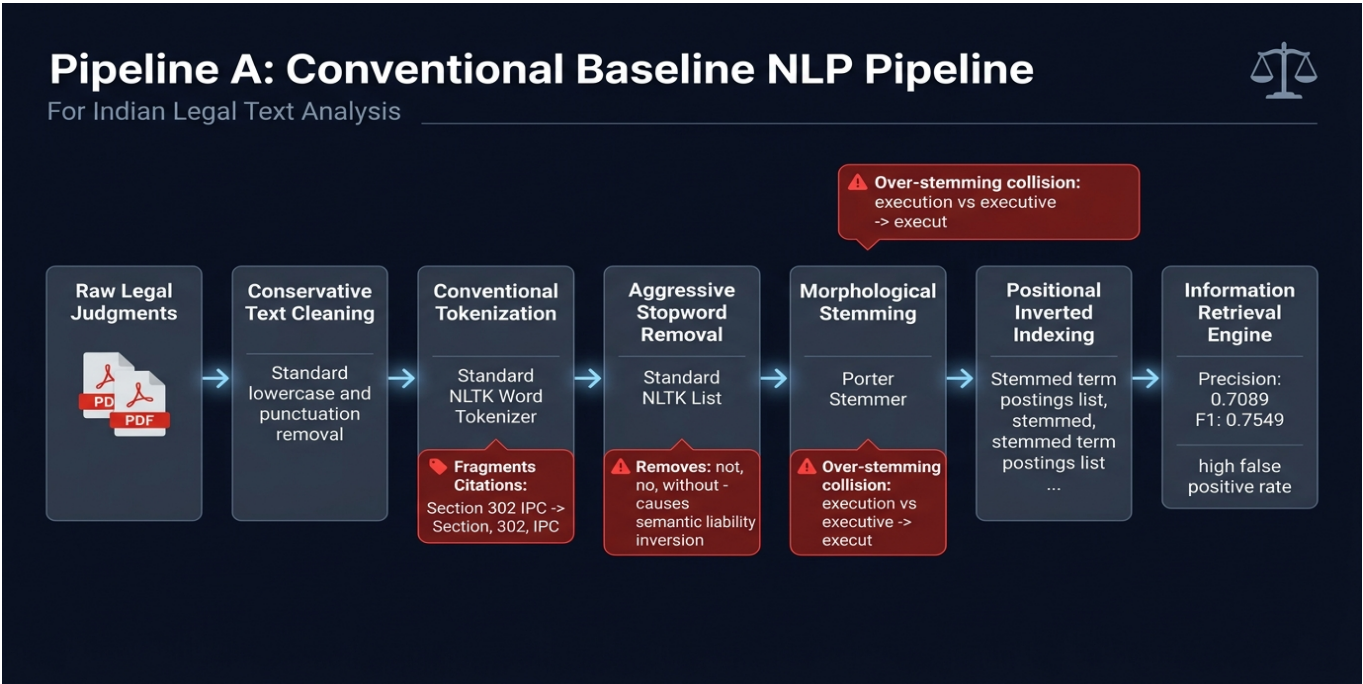
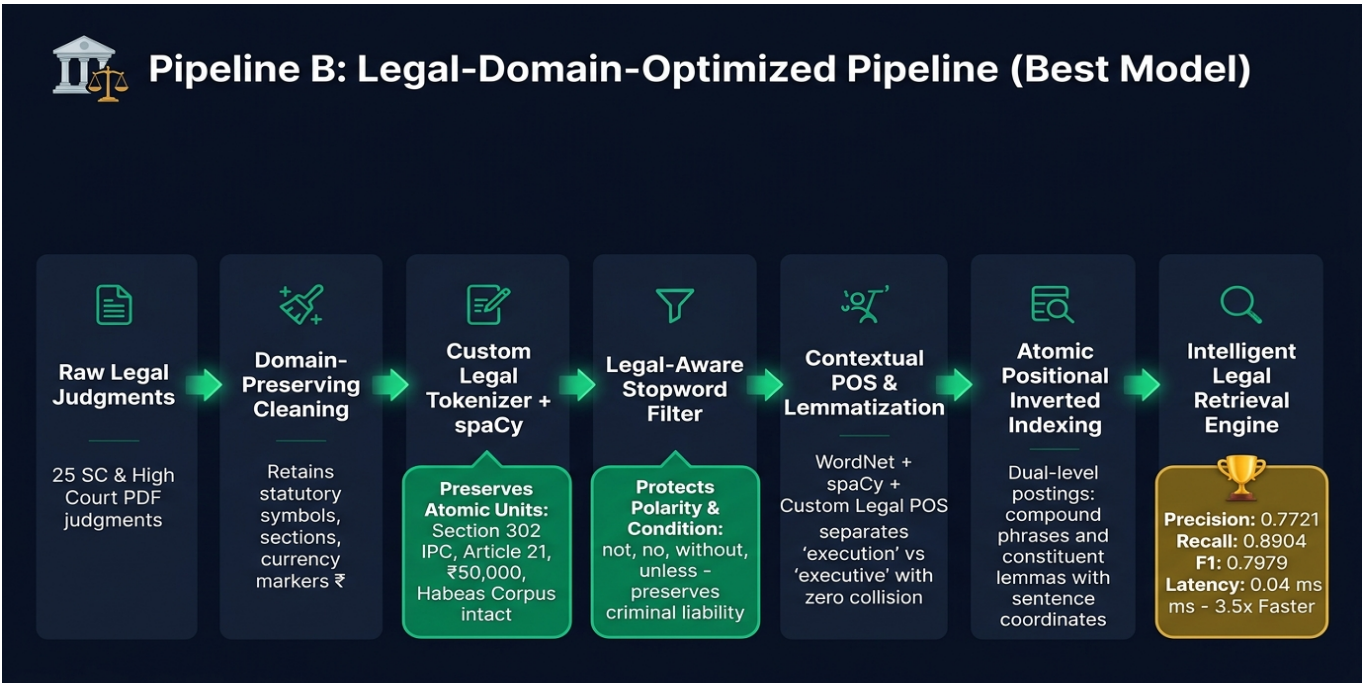


Figure 2: Pipeline B Architecture (Legal-Domain-Optimized — Best Model)



14. Interactive GUI Dashboard & Deployment Architecture

A modern, dark navy legal-tech analytics dashboard was engineered using **Streamlit** across 17 comprehensive functional pages:

- **Corpus Exploration:** High-level KPI telemetry, full-text document viewer with keyword highlighting, lexical density charts.
- **NLP Diagnostics:** Real-time tokenization workbench, stopword comparison, searchable stemming/lemmatization table, POS tagging, interactive document-level NER visualizer, 1-to-5 gram tables, and interactive BPE subword segmenter.
- **Information Retrieval:** Flagship search engine with **live real-time Precision, Recall, and F1 calculation on every query**, academic forensic execution trace (Forensic IR format), interactive ground-truth fine-tuning console, comparative postings table, and Boolean query builder.
- **Cloud Deployment Readiness:** Integrated automated startup bootstrap for NLTK datasets and spaCy models, self-contained dataset path resolution, and full git repository packaging for Streamlit Community Cloud.

15. Limitations, Ethical Considerations & Scholarly References

Limitations & Scalability: While the 25 judgments thoroughly satisfy the assessment requirement, scaling to millions of judgments will benefit from distributed indexing (e.g. Apache Lucene / Elasticsearch) and dense transformer embeddings (LegalBERT).

Ethical Considerations: Legal retrieval tools must serve solely to assist legal research and never replace judicial discretion. Preserving negation ensures algorithmic fairness and prevents prejudicial false positives in criminal cases.

References:

1. Manning, C. D., Raghavan, P., & Schütze, H. (2008). *Introduction to Information Retrieval*. Cambridge University Press.
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5. Chalkidis, I., et al. (2020). *LEGAL-BERT: The Muppets straight out of Law School*. EMNLP Findings.